

Conversational Implicatures: Modelling Relevance Theory Probabilistically

Christoph Unger 

Hendrik Buschmeier 

Faculty of Linguistics and Literary Studies,
Bielefeld University, Bielefeld, Germany
{christoph.unger|hbuschme}@uni-bielefeld.de

Abstract

Recent advances in Bayesian probability theory and its application to cognitive science in combination with the development of a new generation of computational tools and methods for probabilistic computation have led to a ‘probabilistic turn’ in pragmatics and semantics. In particular, the framework of Rational Speech Act theory has been developed to model broadly Gricean accounts of pragmatic phenomena in Bayesian terms, starting with fairly simple reference games and covering ever more complex communicative exchanges such as verbal syllogistic reasoning. This paper explores in which way a similar Bayesian approach might be applied to relevance-theoretic pragmatics (Sperber & Wilson, 1995) by study a paradigmatic pragmatic phenomenon: the communication of implicit meaning by ways of (conversational) implicatures.

1 Introduction

Recent advances in Bayesian probability theory and its application to cognitive science (e.g., Tenenbaum, Griffiths and Kemp 2006; Tenenbaum, Kemp et al. 2011) in combination with the development of a new generation of computational tools and methods for probabilistic computation (Mansinghka 2009; Fierens et al. 2015) have led to what is sometimes called a ‘probabilistic turn’ in pragmatics and semantics (Erk 2022). In particular, the framework of Rational Speech Act theory (Degen 2023) has been developed to model broadly Gricean accounts of pragmatic phenomena in Bayesian terms, starting with fairly simple reference games (Frank and Goodman 2012) and covering ever more complex communicative exchanges such as verbal syllogistic reasoning (Tessler, Tenenbaum and Goodman 2022).

In this paper we explore in which way a similar Bayesian approach might be applied to relevance-theoretic pragmatics (Sperber and Wilson 1995; Clark 2013). To this end, we study the perhaps most paradigmatic pragmatic phenomenon, the communication of implicit meaning by ways of (conversational) implicatures.

Our investigation proceeds according to the following plan: in Section 2, we discuss the phenomenon and main properties of implicatures in informal terms. Section 3 reviews the relevance theoretic account of implicature comprehension and how this is intertwined with explicature comprehension. In Section 4 we explore some of the depths of the theoretical concepts involved and explain how we propose to capture these notions in the probabilistic model we aim to construct. Finally, in Section 5 we

explain in detail a probabilistic model of implicature comprehension implemented in the probabilistic logic programming language ProbLog.

2 Introducing Implicature

Consider the following conversation between Peter and Mary:

- (1) a. *Peter*: Do you want some coffee?
- b. *Mary*: I don't want to drink energizing drinks at this time of day.

Intuitively, what Mary intends to communicate to Peter can be paraphrased as in (2):

- (2) Mary does not want to drink coffee at 6 pm because coffee is an energizing drink and she does not want to drink energizing drinks at 6 pm.

Let us re-order the component conjuncts of this paraphrase of Mary's intended meaning as in (3) and ask about the relationship of each component to Mary's utterance in (1b):

- (3) a. Coffee is an energizing drink.
- b. Mary does not want to drink coffee at this time of day.
- c. Mary does not want to drink energizing drinks at 6 pm.

(3c) is derived from Mary's sentence in (1b) by assigning reference to the pronoun *I* and resolving the time reference of the expression *at this time of day* in the context of the conversational setting. As a result, we can assign truth conditions for the resulting representation (3c), something we can not do for the linguistic representation (1b) (the sentence uttered by Mary). Notice, however, that (3c) has a close relationship to the logical form of (1b): it is derived from the logical blueprint given by the sentence through contextual enrichment and reference resolution. Let us call this process of contextually supplementing logical forms to the points where the resulting representations carries truth conditions the *development* of the logical form of the utterances (Sperber and Wilson 1995). Representations that the audience derives as a result of such a process of the development of the logical form of an utterance are *explicitly* communicated and are called *explicatures* (Sperber and Wilson 1995).

However, (3c) alone does not suffice to answer Peter's question. Yet Mary ostensibly presents her utterance of (1b) as an answer to Peter's question. In order to explain how Mary could have intended her utterance to answer Peter's question, Peter has to supply the assumption (3a) as a contextual premise. From this contextual premise and the representation (3c) Peter can conclude (3b) that *Mary does not want to drink coffee at 6 pm*. Notice that neither (3a) nor (3b) are derived by further enriching the logical form of Mary's utterance; rather, (3a) is an independent proposition that together with the proposition conveyed by Mary's utterance yields the conclusion (or in other words, contextual implication) (3b). Implicit premises and conclusions that need to be contextually supplied in order for the audience to comprehend the speaker's meaning, but are not the result of the development of a logical form, are fully *implicitly* communicated and are called *implicatures*.

3 Comprehending implicatures

3.1 Verbal communication works by ostension

What is Mary achieving when she utters (1b)? As we have seen, she has not encoded all aspects of the meaning she wanted to convey to Peter into her utterance. Peter has to supplement the logical form he can decode from the linguistic form of Mary's utterance in two ways:

- (4) a. Develop the logical form of the utterance into a proposition by reference resolution of indexicals using contextual evidence.
- b. Selecting a suitable implicit premise as a contextual assumption for deriving an implicit conclusion that answers Peter's question (and thereby allows him 'to see the point' of Mary's utterance).

What Mary has done by uttering (1b) is that she has provided Peter with partial evidence for inferring her intended meaning. In other words, comprehension is an exercise in intention attribution.

That comprehending verbal communication is essentially an act of intention attribution was one of the ground breaking insights of the philosopher Paul Grice, discussed in his seminal paper *Meaning* (Grice 1957). Sperber and Wilson (1995) have taken up these ideas and refined them by arguing that in verbal communication the communicator makes overt a layered set of intentions as follows:

- (5) a. *Informative intention*: to make manifest or more manifest to the audience a set of assumptions **I**. (Sperber and Wilson 1995, p. 58)
- b. *Communicative intention*: to make it mutually manifest to audience and communicator that the communicator has this informative intention. (Sperber and Wilson 1995, p. 61)

Central in these definitions is the concept of the *manifestness* of a piece of information (Sperber and Wilson 1995, pp. 38–46). We quote their definition from a later paper in which the authors elaborate further on the impact of the notion of *manifestness*:

- (6) A proposition¹ is manifest to an individual at a given time to the extent that he is likely to some positive degree to entertain it and accept it as true. (Sperber and Wilson 2015, p. 134)

The manifestness of a piece of information is influenced by two factors: the epistemic status of the piece of information (how likely it is to be accepted as true), and its salience or accessibility (how likely it is to be considered at all). Manifestness is the joint influence of these factors and is a measure for the probability that the given piece of information will influence an individual's beliefs or decisions (Sperber and Wilson 2015, p. 134).

The communicator's informative intention was defined as the intention to increase the manifestness of a set of assumptions to an audience. This set of assumptions is the

¹Directly after presenting this definition, Sperber and Wilson make the following comment: "In *Relevance*, we talked not of propositions but of assumptions as being manifest. Either term will do, and so would 'pieces of information'. What we are talking about are things that can be true or false and that, when they are true, are facts." (Sperber and Wilson 2015, p. 134)